

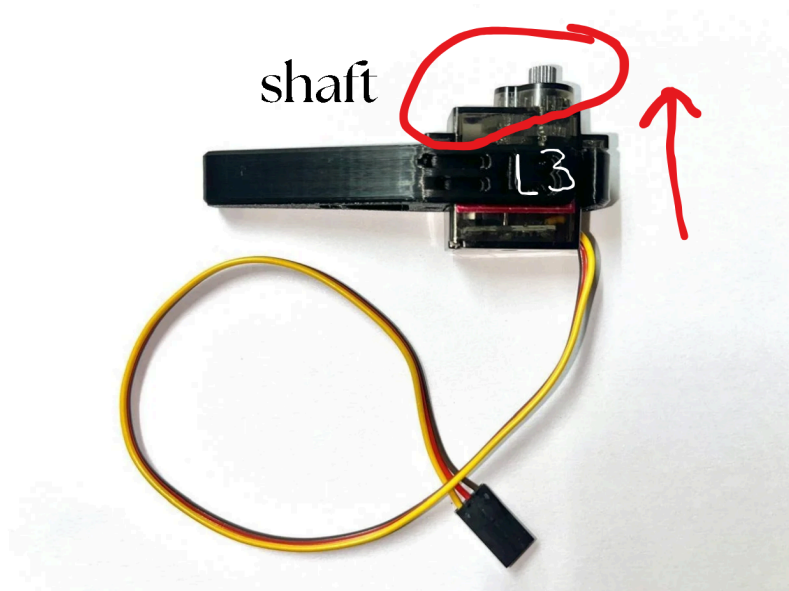
Single Leg Assembly Guide

Part 1: Attach the Servo Motor to the Leg

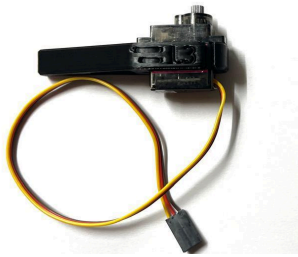
You are mounting the servo motor onto the leg so that when the servo rotates, it moves the tibia.

Step 1. Hold the tibia so the label faces you and is readable (the letter & number are not upside down).

Step 2. Hold the servo motor so the servo shaft faces up. The servo shaft is the silver, round, ridged, cylindrical piece at the top of the servo motor that rotates.

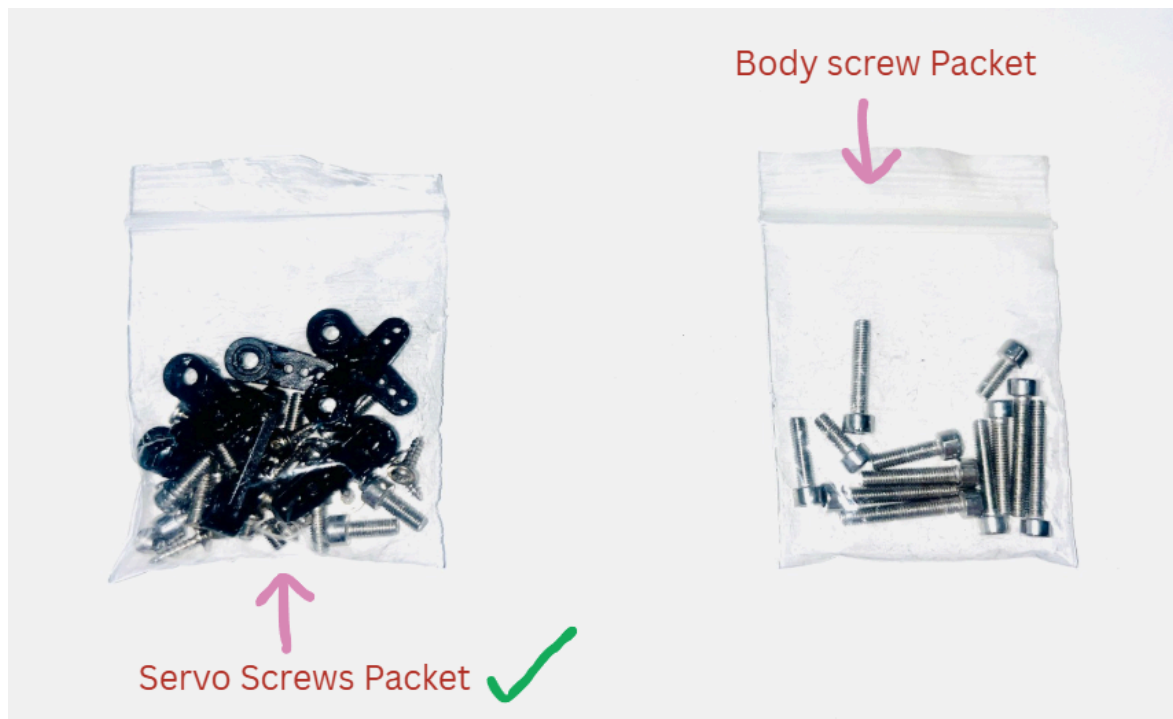


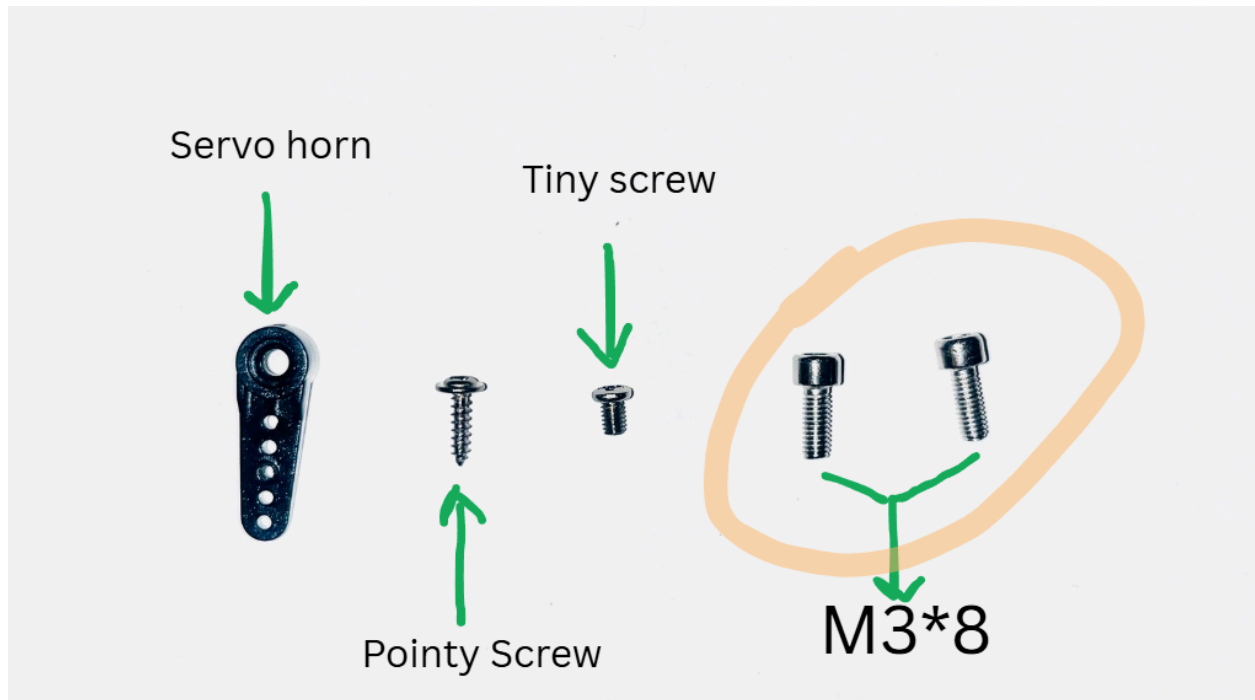
Step 3. Insert the servo motor into the rectangular cutout on the tibia, positioning it so the shaft is near the top of the leg and pointing upward, above the L3 label. The servo motor should sit flat against the tibia once it has been pushed in. Do not force it. If it does not sit flat, check the orientation and call an instructor.



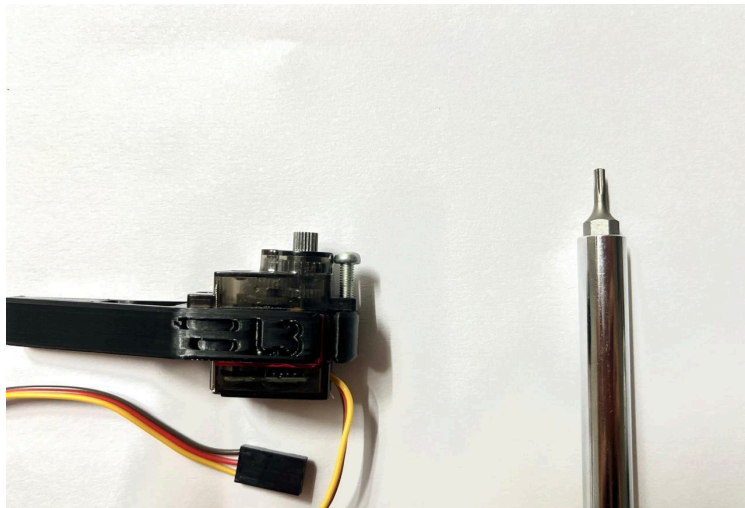
Step 4. Insert the H2.5 screwdriver tip into the screwdriver body. You can identify a tip as H2.5 by looking at the engraving on its side.

Step 5. You will have two packets. Use the servo screws packet. Take two 3x8 mm screws (hex screwhead)



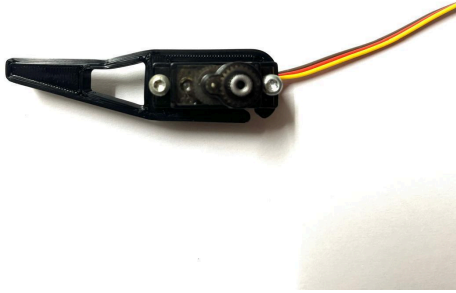


Step 6. Place one screw into the screw hole on one side of the servo motor and the tibia together. Turn the screwdriver right-ward/clockwise (tighten) slowly until the screwhead touches the servo. Do not overtighten as it may damage your motor and/or tibia.



Step 7. Repeat with the second screw on the other side of the motor.

Step 8. The servo motor is now firmly attached to the tibia. The top view of how the tibia should look is given below. Set the leg aside.



Q: Why Do We Set the Servo to 90 Degrees Before Attaching the Horn?

A servo motor can rotate from 0 degrees to 180 degrees. The center position is 90 degrees.

For our quadruped robot (a four-legged walking robot), each tibia needs to move equally in both directions from the center. If the servo horn is attached while the servo is at the wrong angle, the leg will not be centered and will be able to move further in one direction than the other. By setting the servo to exactly 90 degrees before attaching the horn, we ensure the horn is physically centered so the leg can move the same amount in both directions.

Part 2: Connecting the Servo Motor to the ESP32

You will now connect the servo motor to the XIAO ESP32-C3 using three male-to-female jumper wires.



What the pins mean:

- **GND** — Ground. This is the negative reference point for all electrical signals. Every electronic component in a circuit must share a common ground to work correctly.
- **VUSB** — This pin provides power drawn from your laptop through the USB/C to C cable. The servo motor needs this power to move.
- **Do** — This is a digital input/output pin on the ESP32. Do means Digital Pin 0. It sends the signal to the servo motor telling it what angle to move to. Your ESP32 has many pins labelled Do, D1, D2, and so on, each one can send or receive signals independently.

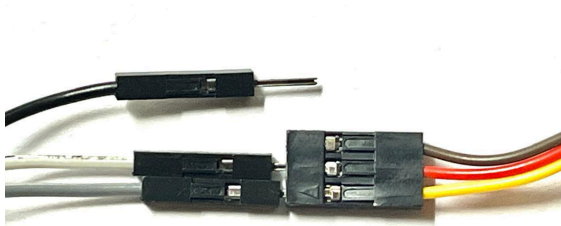


Step 1. Look at the servo motor. It has three wires coming out of it:

- Brown or black wire — this is Ground
- Orange/Red wire — this is Power
- Yellow wire — this is Signal

Step 2. Take one male-to-female jumper wire. Plug the female socket end onto the brown or black wire of the servo. Plug the male pin end into the GND pin on the ESP32.

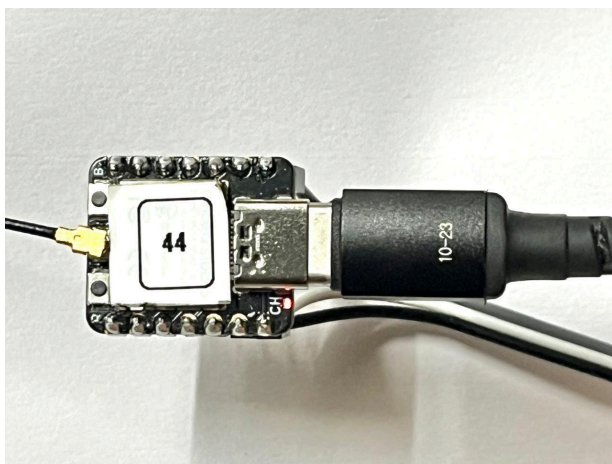
Step 3. Take a second male-to-female jumper wire. Plug the female socket end onto the red/orange wire of the servo. Plug the male pin end into the VUSB pin on the ESP32.



Step 4. Take a third male-to-female jumper wire. Plug the female socket end onto the yellow wire of the servo. Plug the male pin end into the D-0 pin on the ESP32.

Why three wires? Every component needs power (+), ground (-), and a way to receive instructions (signal).

Step 5. Plug the USB-C end of your cable into the ESP32. The red light on the ESP32 should blink or turn on, confirming it has power.



Check that the baud rate is set to **115200**. If it is not, change it to 115200. The baud rate is how many communication symbols are sent & received each second.

Step 8. In the text input box at the bottom of the Serial Monitor panel, type `90` and press Enter. You will see the message `-> Moved to 90 degrees`. The servo is confirmed at center position. When the servo moves, you will see the shaft rotating and hear a noise.

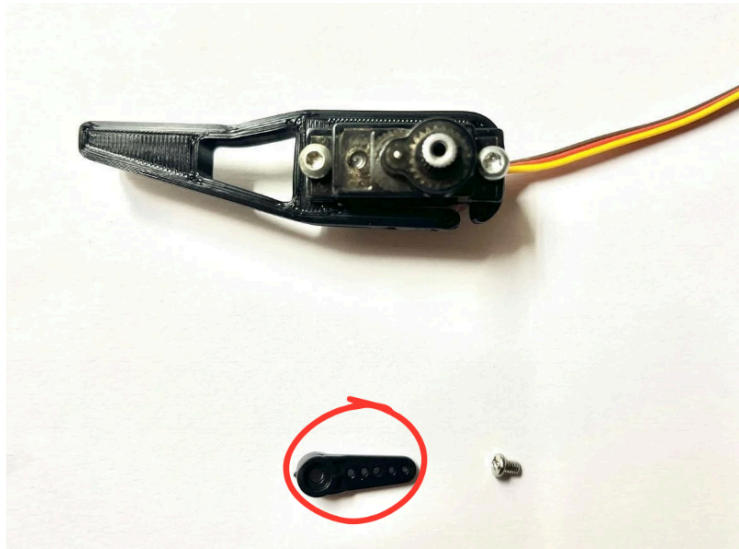
Caution: If your servo is continuously making a noise for more than 5-10 seconds/getting heated (feels warm/hot when you touch the body), immediately disconnect the USB/C cable from your laptop and call an instructor.

Step 9. You may now also type `0` and press Enter to see the servo move to one extreme, and `180` and press Enter to see it move to the other extreme (you can enter any angle for that matter!). Type `90` again to return to center before the next step.

Step 10. Unplug the USB cable from your laptop. The servo will hold its last position.

Part 6 Attach the Servo Horn

Step 1. Take the servo horn from the servo horn packet. It is a plastic piece with an arm extending outward. It has been circled in the image below.

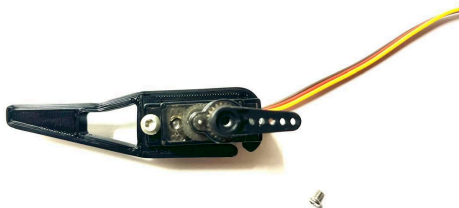


Use this for
screwing
servo horn on
top of shaft



Look at the top of the servo motor. The output shaft has ridges around it. The servo horn has a matching ridged hole in its center. These ridges are designed to lock together so the horn cannot slip.

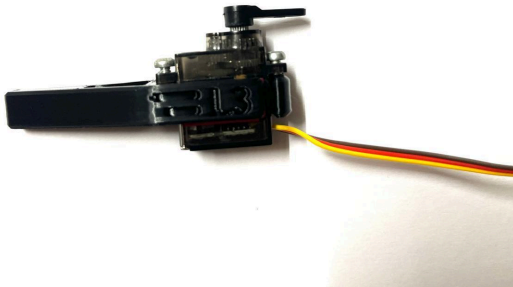
Step 3. Hold the tibia so the label faces you. Orient the servo horn so that one arm points straight outward away from the leg, at 180 degrees relative to the tibia body. This is the correct horn position (with respect to how it is going to be positioned in the main frame body).



Step 4. Press the horn down firmly onto the output shaft. You should feel it click into place. Do not rotate it once it is on, simply press straight down.

Step 5. Insert the PH01 screwdriver tip into the screwdriver body.

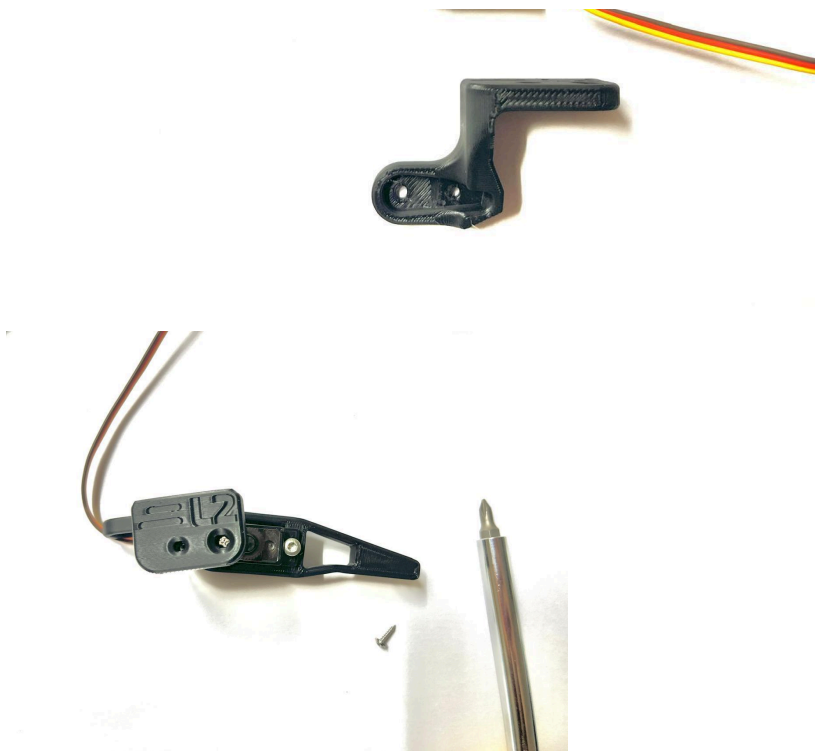
Step 6. Take the small servo horn screw from the servo horn packet. This screw is smaller than the 3x8mm screws, it is specifically sized for the center hole of the servo horn. It is not pointy.



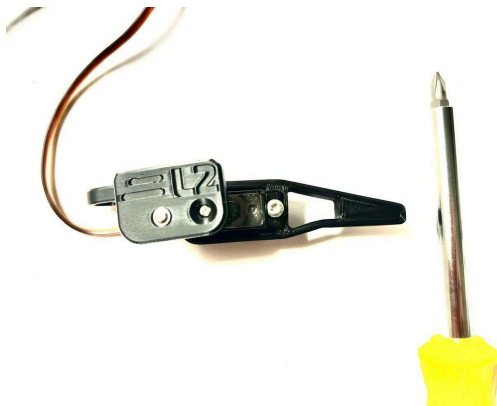
Step 7. Place this screw into the hole of the servo horn. Turn the screwdriver clockwise until the screw is inside the cavity. Do not overtighten as this can crack the plastic horn.

Part 7 Attach the Hip Joint

Step 1. Take the femur. Look at it carefully. It has two cavities (hollow openings) on it. One cavity is for the servo horn arm to fit into, and the other is for the body/frame. The cavity for the body frame has the label on its flip side. For the horn, use the other cavity. Slide the horn arm into this cavity. It should slide into this cavity perfectly and not be loose.



Step 4. Take the pointy screw from the servo horn packet. This screw secures the femur with the rest of the leg.



Step 5. Using the PH01 screwdriver tip, insert this screw through the hip joint and into the leg piece. Turn clockwise to tighten it.